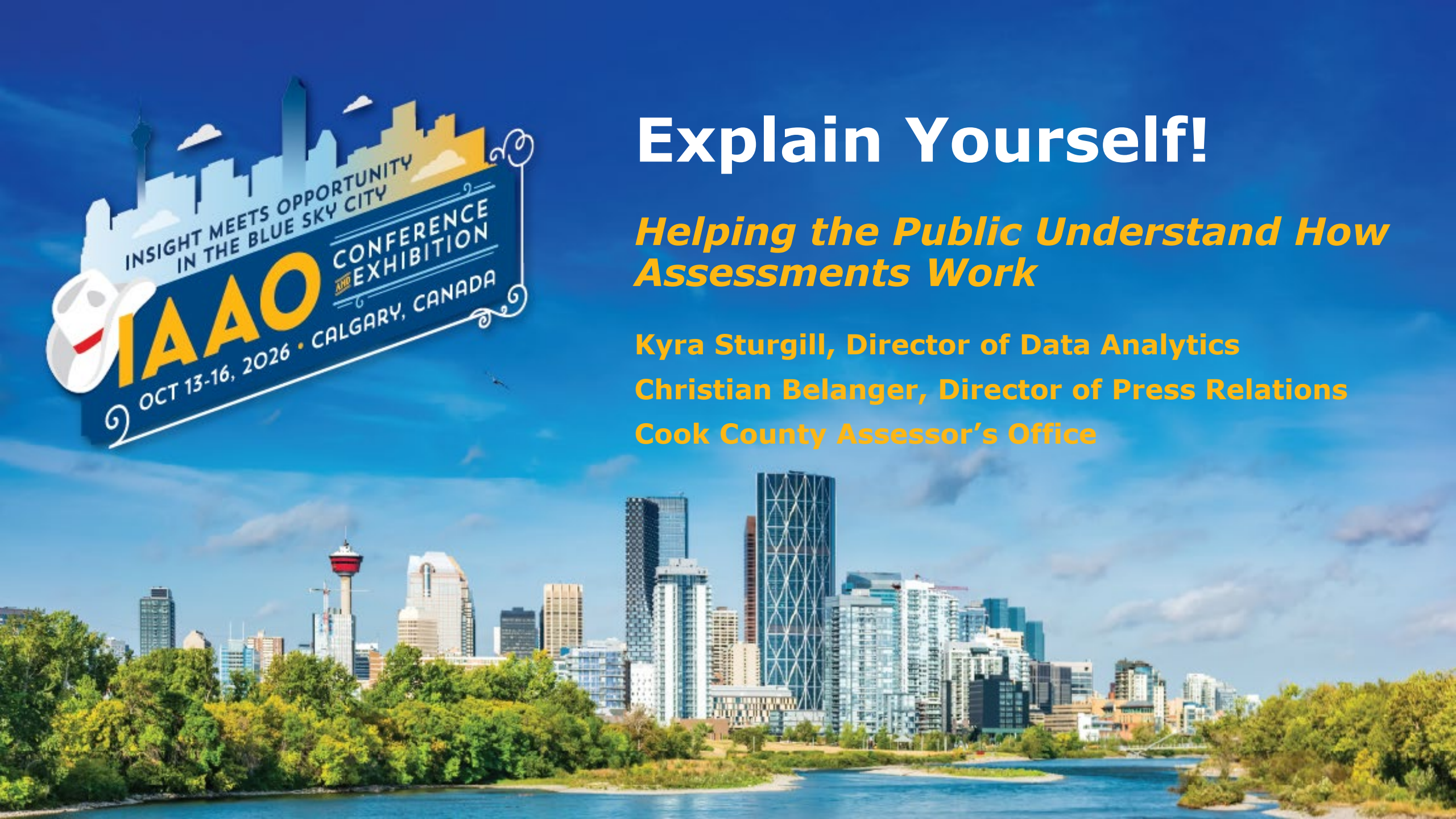


Explain Yourself!

Helping the Public Understand How Assessments Work

Kyra Sturgill, Director of Data Analytics

Christian Belanger, Director of Press Relations

Cook County Assessor's Office

The context

- **1.2 million residential parcels in Cook County**
- **Triennial reassessments**
  - Chicago, North, South
- **Poster child for residential regressivity**

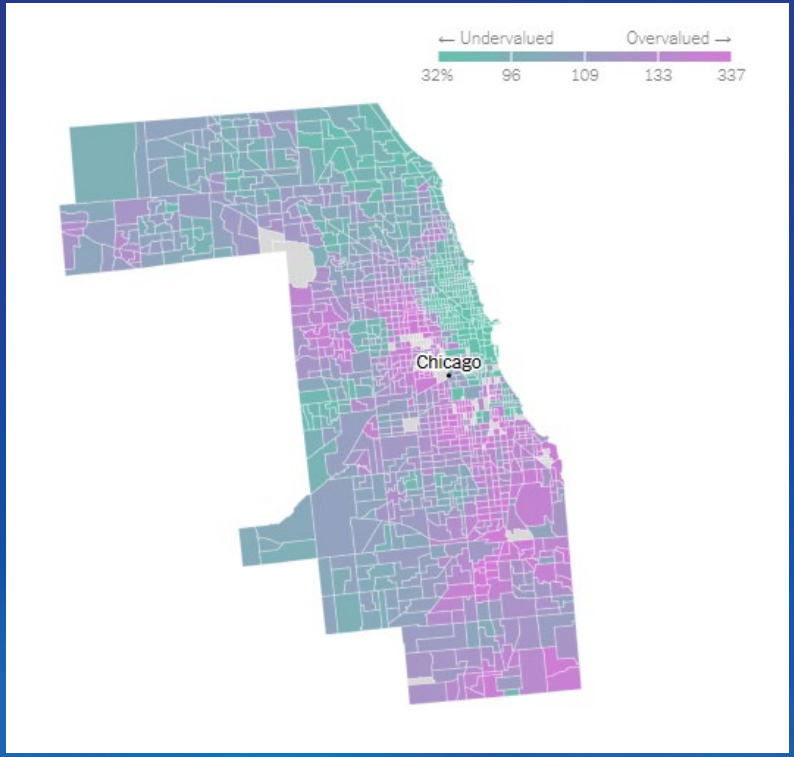


Figure 2: Net Tax Shift by Sale Price Decile

The Tax Divide

ProPublica Illinois a Finalist for the Pulitzer Prize for Local Reporting

Grotto began his reporting with a focus on residential assessments. He gathered data on single-family homes and apartment buildings to understand how average homeowners were affected. Assessment officials released the information he requested, never thinking that a journalist could process the reams of data or understand the complex regression models used to value residential property. But with the help of his sources, Grotto decoded the models and showed how the system treated poorer homeowners unfairly. The assessment system, it turned out, reflects the racial and socioeconomic divisions that have long made Chicago one of the most segregated cities in America. Through neighborhood anecdotes, the series showed how the assessment system awards breaks to the already-privileged residents of wealthy and mostly white parts of the county, while laying new burdens on working class areas where most residents are black or Latino.



Making progress

- **Incoming assessor in 2018 establishes Data Science department**
- **New residential valuations model**
- **Public documentation**

Pinned

model-res-avm Public

Automated valuation model for all class 200 residential properties in Cook County (except vacant land and condos)

R ☆ 73 20

model-condo-avm Public

Automated valuation model for all class 299 and 399 residential condominiums in Cook County

R ☆ 14 6

model-sales-val Public

Heuristics for detecting outlier and non-arms-length sales

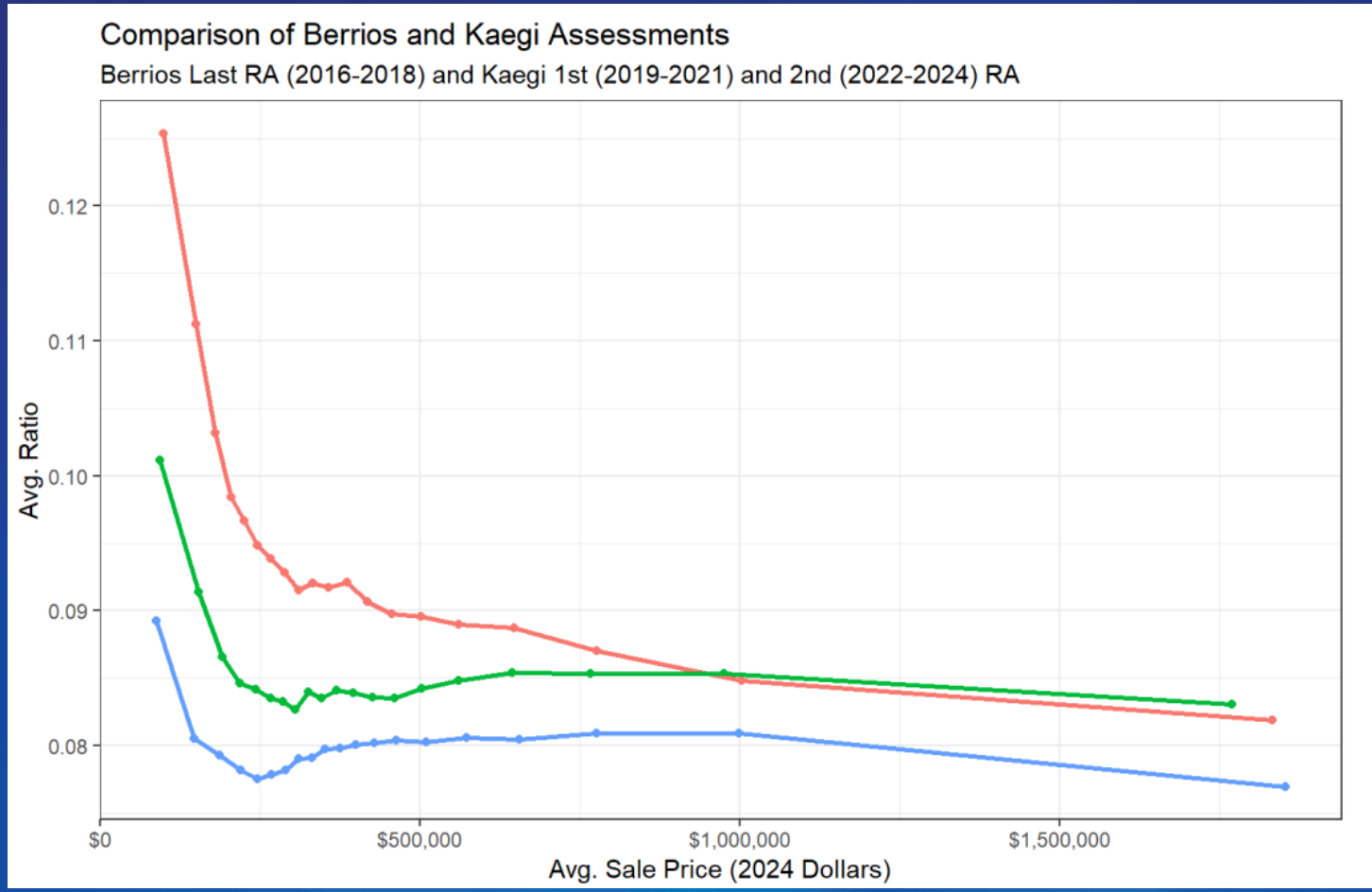
Python ☆ 2 2

NEWS > POLITICS

Study finds Kaegi has improved fairness in Cook County property assessments

"The Kaegi administration has made substantial progress in improving the fairness of residential assessments. By all measures, ***regressivity—the tendency for lower priced properties to be assessed at higher rates, relative to their actual sale price, than higher priced properties— has been dramatically reduced****..."*

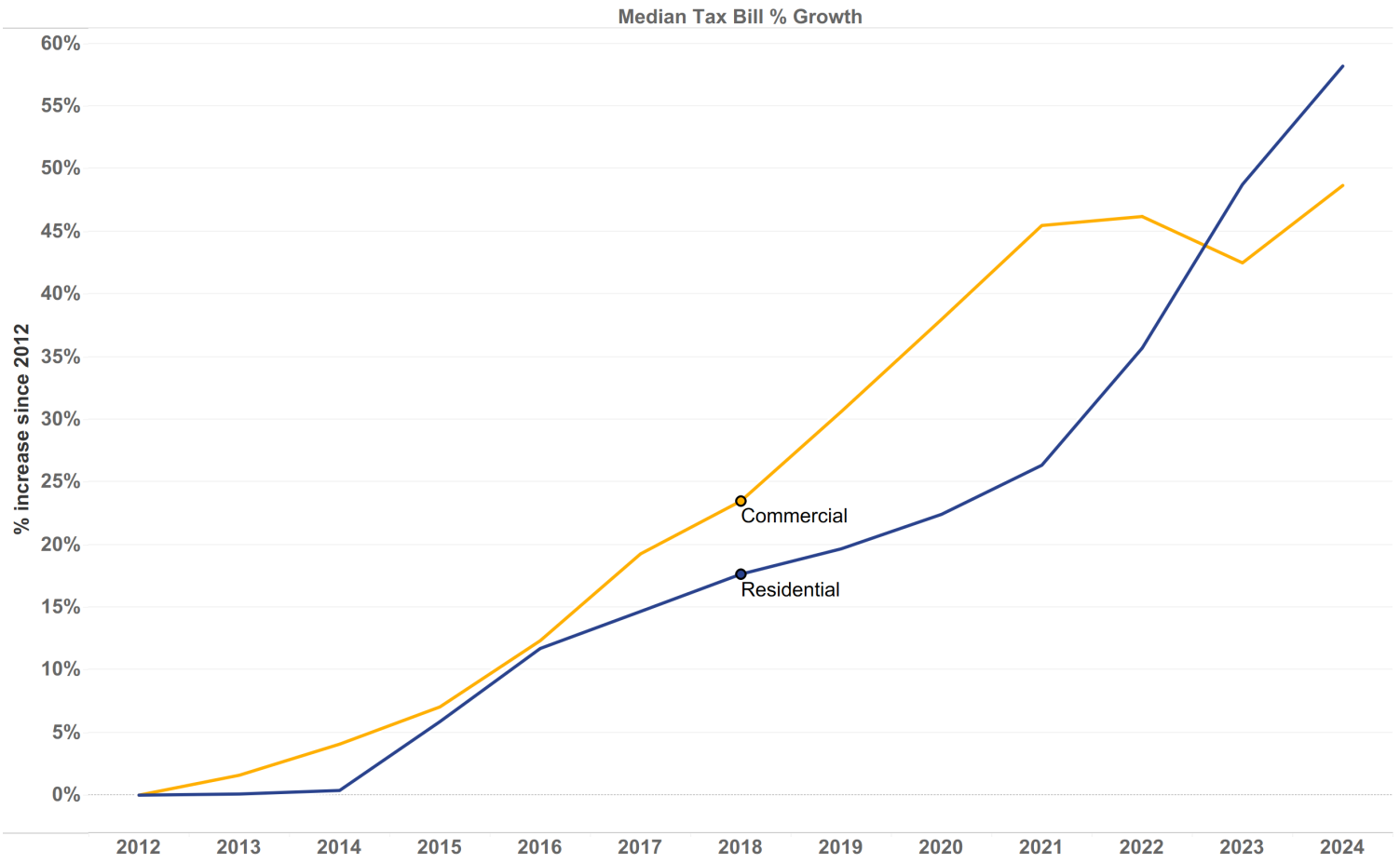
"These improvements have resulted in ***over a billion dollars in aggregate tax savings*** *for owners of middle and lower priced properties..."*



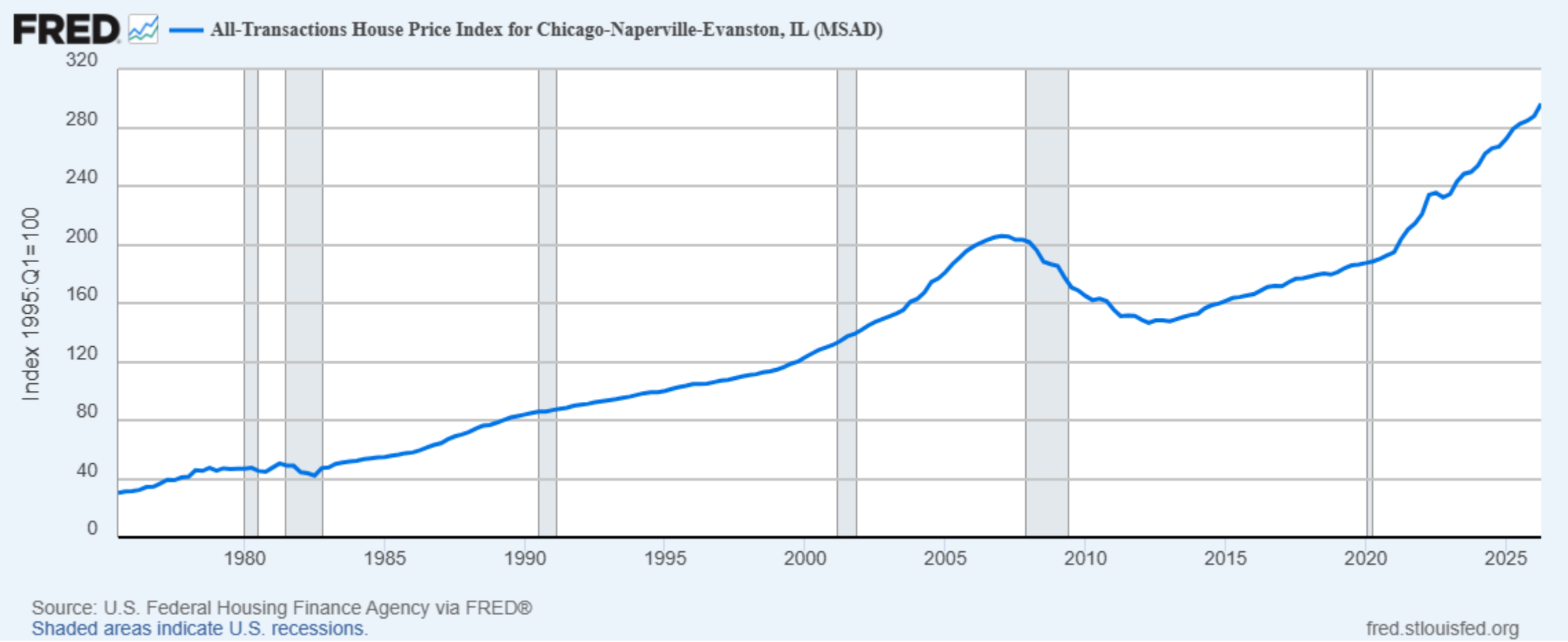
Cook County property taxes are high…

…and getting higher

- Cook County median effective tax rate is 2.14%, higher than the national median of 1%
- Median tax bills for commercial and residential properties have increased since 2012

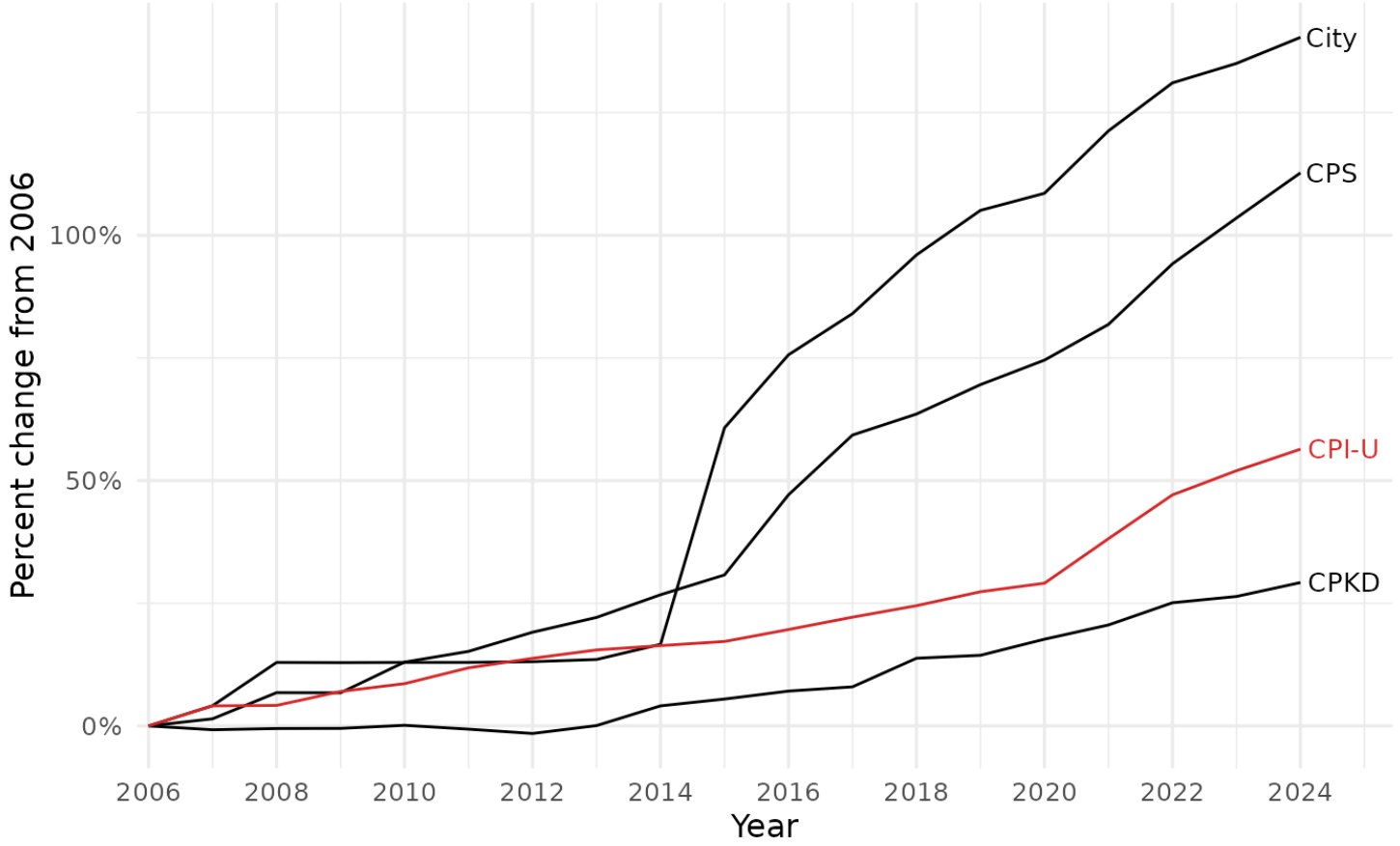


Unprecedented times



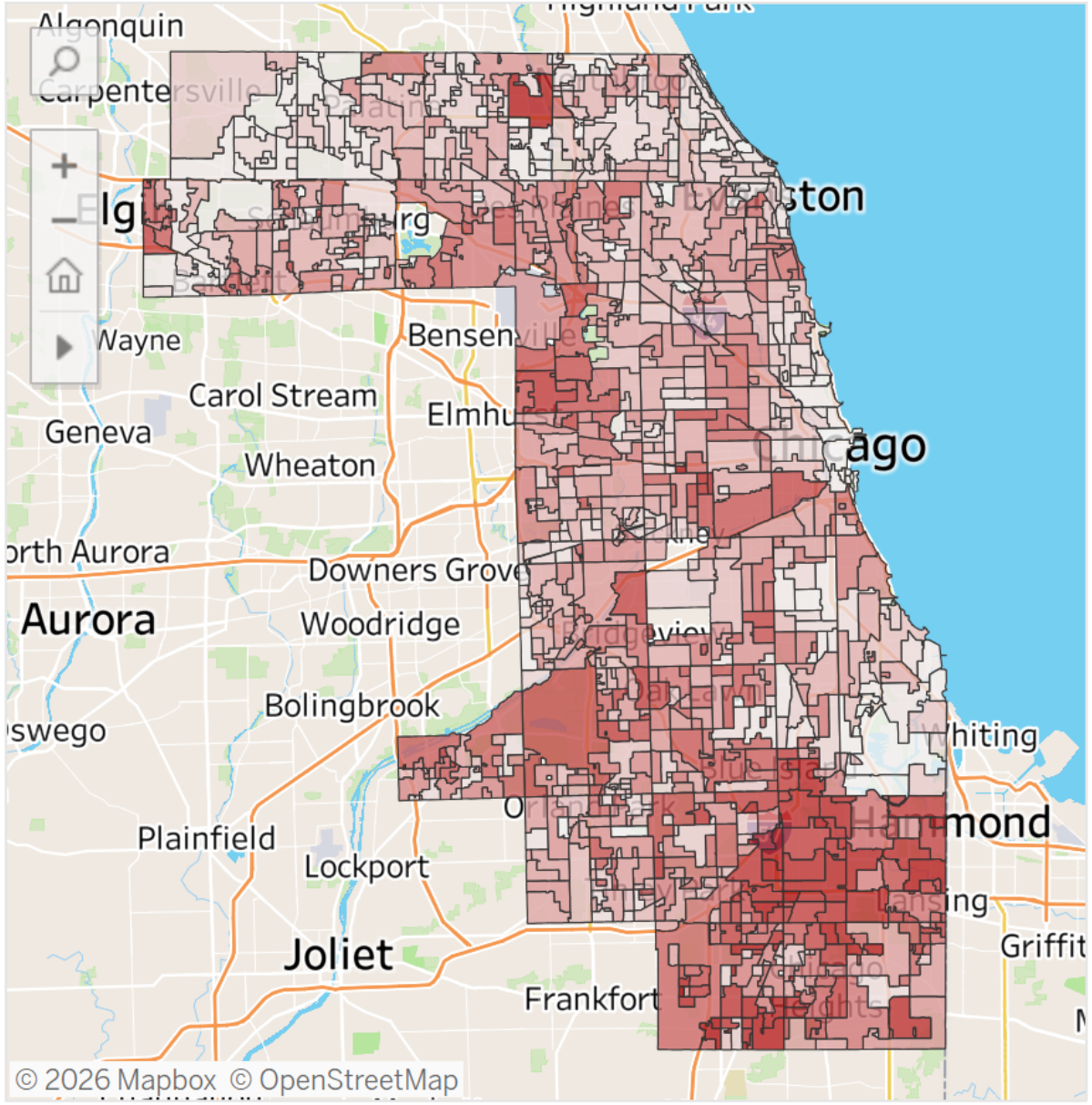
Factors beyond CCAO's control

Taxing agencies increasing levy beyond the rate of inflation



236,841 homeowners

had a **property tax bill spike** between Tax Years 2021-2023.



Nearly 25% of homeowners in Cook County experienced a tax spike* between 2021 and 2023

Median tax spike among these properties was around $1,700

*a one-year **tax spike** in 2020-2021, 2021-2022, or 2022-2023 of more than **25% and $500**, with a max market value of **$1,000,000.**

Universe of PINs: this dashboard calculates tax spikes among *homeowners*, meaning residential (class 2) PINs that have received at least one homestead exemption, and had the same exemption(s) from 2020-2021, 2021-2022, and/or 2022-2023.

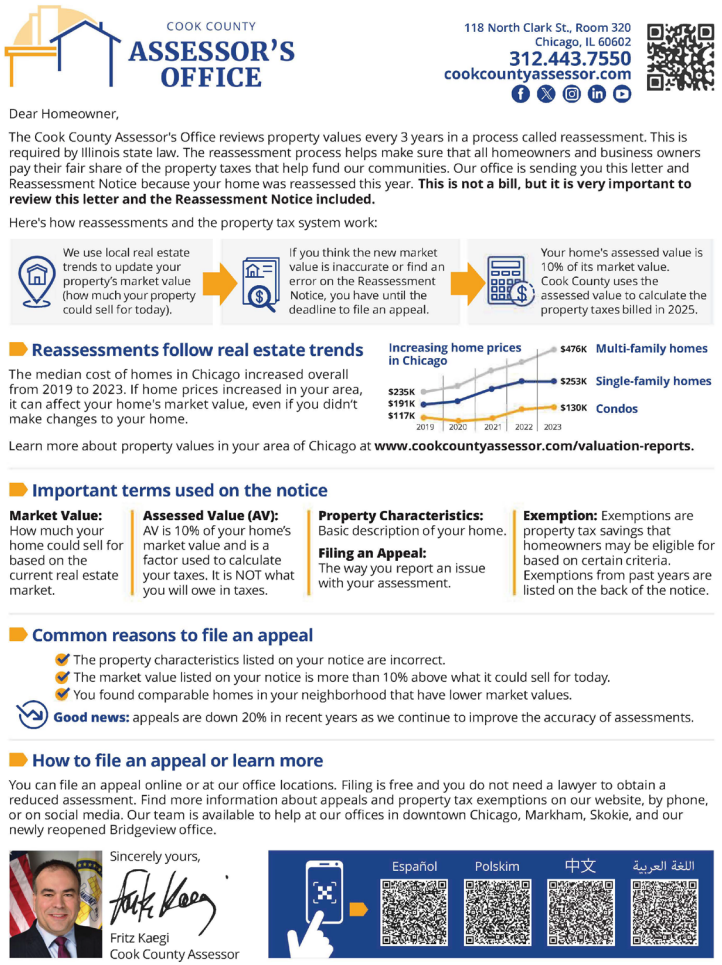
Preparing for 2024 Tax Bills (in 2025)

2024 reassessments for Chicago

- **Notices (just redesigned!) are mailed in Spring/Summer/Fall of 2024**

12-18 months later…

- **The 2nd installment tax bill that reflects updated 2024 assessed values is mailed in Fall 2025**



COOK COUNTY ASSESSOR'S OFFICE

118 North Clark St., Room 320
Chicago, IL 60602
312.443.7550
cookcountyassessor.com

Dear Homeowner,

The Cook County Assessor's Office reviews property values every 3 years in a process called reassessment. This is required by Illinois state law. The reassessment process helps make sure that all homeowners and business owners pay their fair share of the property taxes that help fund our communities. Our office is sending you this letter and Reassessment Notice because your home was reassessed this year. **This is not a bill, but it is very important to review this letter and the Reassessment Notice included.**

Here's how reassessments and the property tax system work:

We use local real estate trends to update your property's market value (how much your property could sell for today).

If you think the new market value is inaccurate or find an error on the Reassessment Notice, you have until the deadline to file an appeal.

Your home's assessed value is 10% of its market value. Cook County uses the assessed value to calculate the property taxes billed in 2025.

Reassessments follow real estate trends

The median cost of homes in Chicago increased overall from 2019 to 2023. If home prices increased in your area, it can affect your home's market value, even if you didn't make changes to your home.

Increasing home prices in Chicago

$476K Multi-family homes
$253K Single-family homes
$130K Condos
$235K
$191K
$117K
2019 2020 2021 2022 2023

Learn more about property values in your area of Chicago at **www.cookcountyassessor.com/valuation-reports.**

Important terms used on the notice

Market Value: How much your home could sell for based on the current real estate market.

Assessed Value (AV): AV is 10% of your home's market value and is a factor used to calculate your taxes. It is NOT what you will owe in taxes.

Property Characteristics: Basic description of your home.

Filing an Appeal: The way you report an issue with your assessment.

Exemption: Exemptions are property tax savings that homeowners may be eligible for based on certain criteria. Exemptions from past years are listed on the back of the notice.

Common reasons to file an appeal

- The property characteristics listed on your notice are incorrect.
- The market value listed on your notice is more than 10% above what it could sell for today.
- You found comparable homes in your neighborhood that have lower market values.

Good news: appeals are down 20% in recent years as we continue to improve the accuracy of assessments.

How to file an appeal or learn more

You can file an appeal online or at our office locations. Filing is free and you do not need a lawyer to obtain a reduced assessment. Find more information about appeals and property tax exemptions on our website, by phone, or on social media. Our team is available to help at our offices in downtown Chicago, Markham, Skokie, and our newly reopened Bridgeview office.

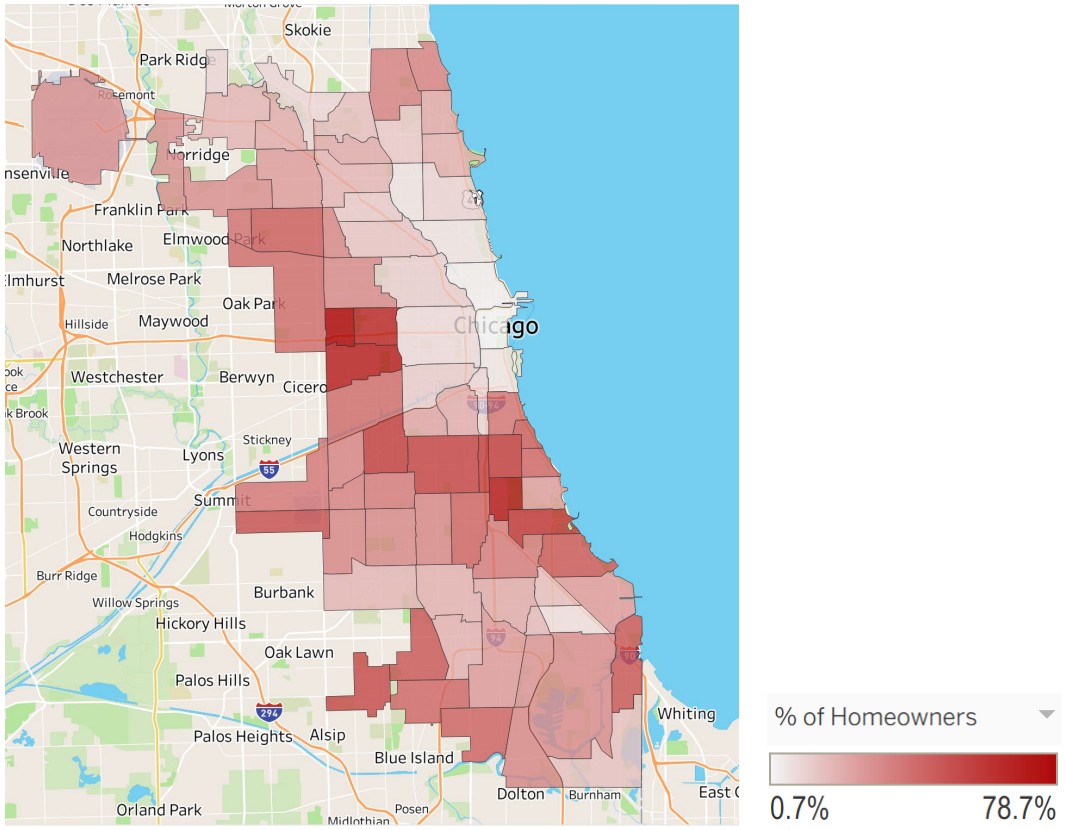
Sincerely yours,

Fritz Kaegi
Cook County Assessor

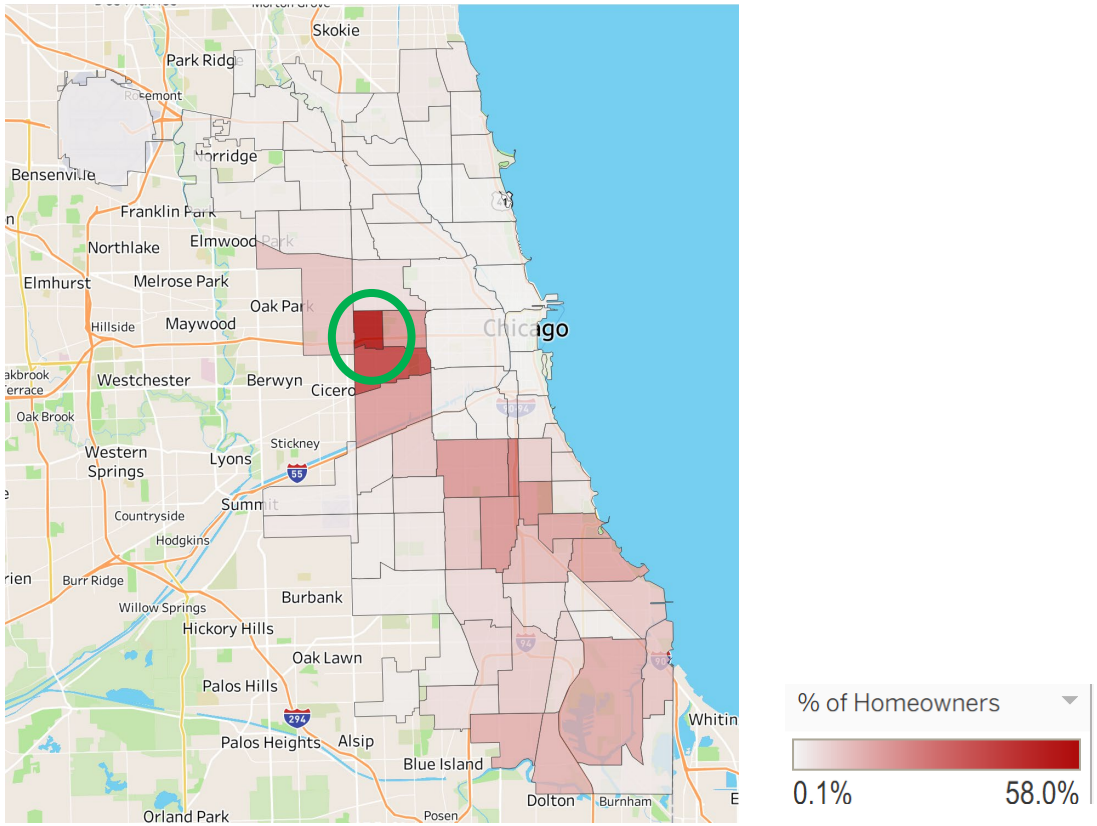
Español Polskim 中文 اللغة العربية

AV increases in 2024 would lead to tax spikes for Chicago

Tax bill increase of at least 25% & $500 for homes valued less than $1M:
82,789 (24%) Chicago homeowners



Tax bill increase of at least 100% & $1000 for homes valued less than $1M:
12,620 (4%) Chicago homeowners



Journalists also take note

Chicago Tribune

Investigations | Brace for impact: Tax hikes loom for South,...

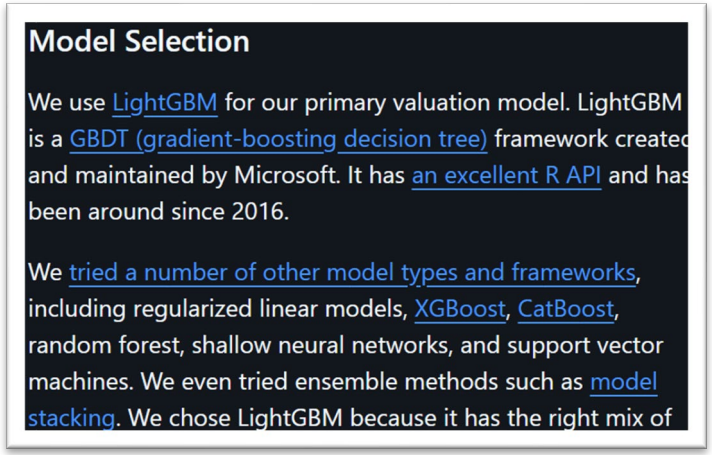
Brace for impact: Tax hikes loom for South, West Side homeowners



“Residential properties throughout the South and West sides, like these shown on Sept. 18, 2025 on West End Avenue in West Garfield Park, had their property assessments double or even triple last year. Those higher assessments will likely help trigger steep hikes in property tax bills for homeowners in the neighborhoods.”

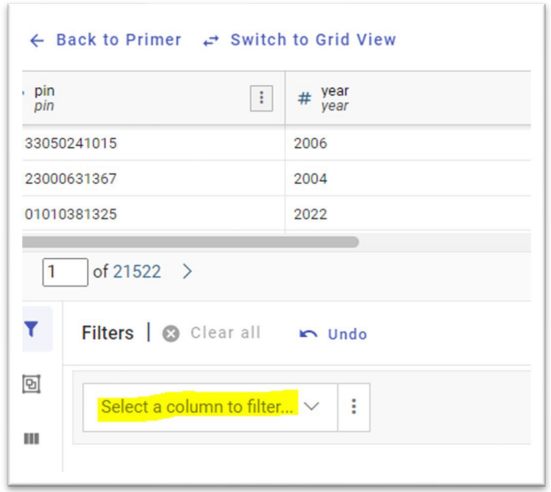
Transparency, Phase 1

27 pages of **model documentation** on github. (And the actual code!)



Requires coding experience and ML knowledge

Complete datasets such as sales and AVs published on **Open Data**



A lot of data available, but requires effort to analyze and interpret

We publish **reporting dashboards** containing aggregate stats about assessments and real estate trends

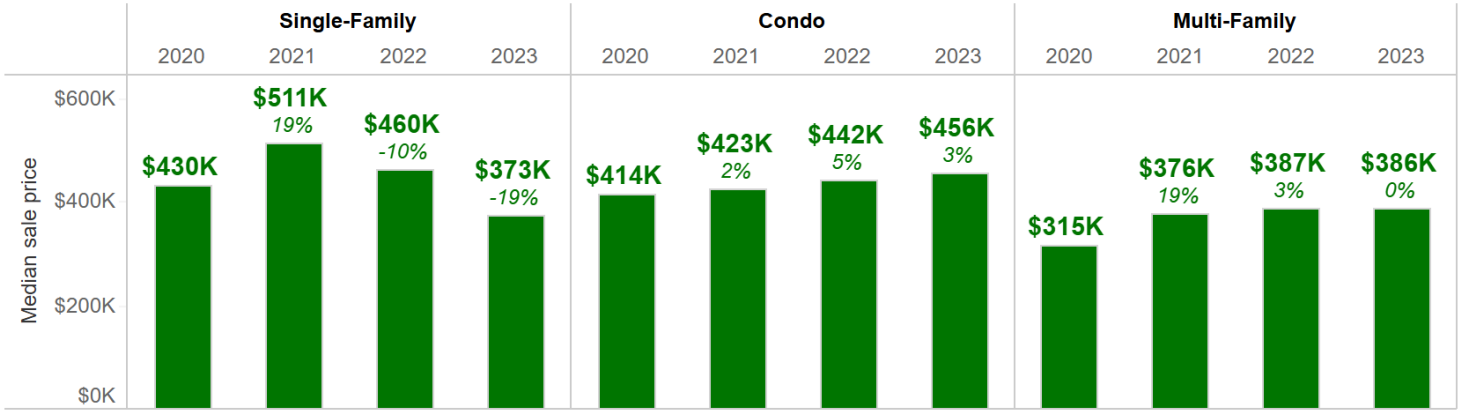


CCAO defined neighborhoods and townships are less familiar geographies

Township and neighborhood reporting

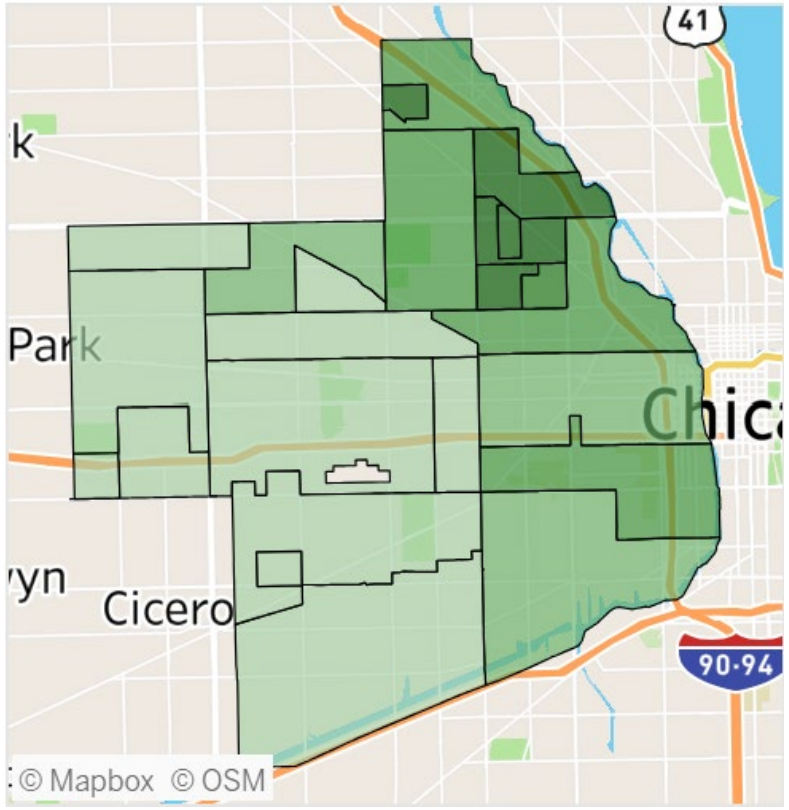
West Chicago Township

Real estate market trends in overall township median sale prices:



Median Assessor's estimated market values in 2024:

Single-Family	Condo	Multi-Family
$256K	**$399K**	**$353K**
16% change from 2021 median	*11% change from 2021 median*	*23% change from 2021 median*

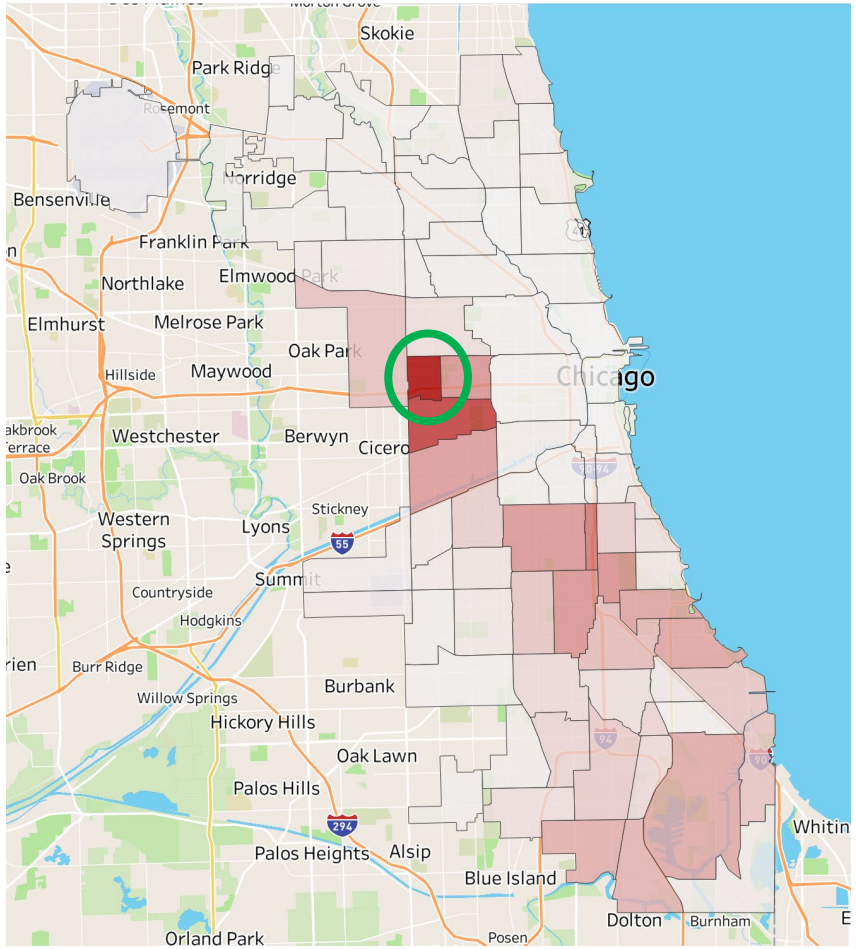


Two-flat in West Garfield Park



2023 Est. FMV: $100,000

2024 Est. FMV: $270,000



Median Sale Price in West Garfield Park, Multifamily Properties

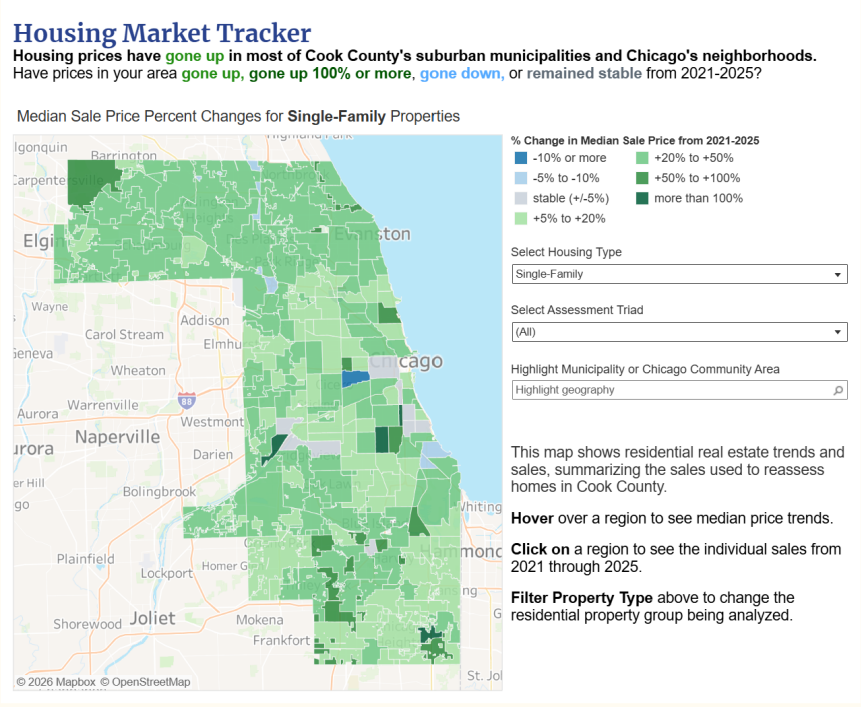
Since 2015, the median sale price has increased over 400%



Transparency, Phase 2: "Explain Yourself" suite of tools

Cook County Housing Market Tracker

- Comprehensive look at all residential sales from the last five years



Home Value Report

- Identifies most significant sales contributing to estimated value of a specific property

Characteristics for Top 5 Most Significant Sales

"Characteristics" are data points about a home that help the model compare it to recent sales. The table below shows all of the characteristics that the model used to estimate your home value alongside the characteristics for the top five sales.

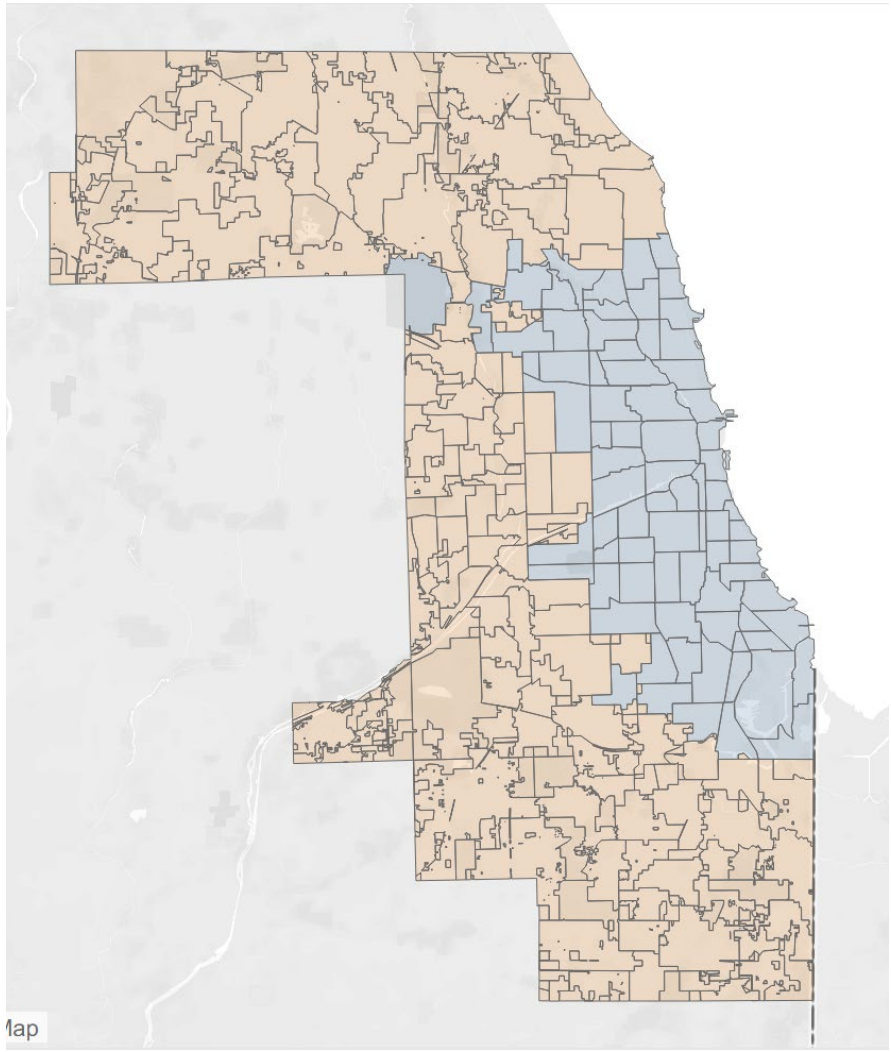
Each characteristic has a score indicating how important it was for the model's estimate. One star (★) indicates that a characteristic was not important, while four stars (★★★★) indicates that a characteristic was very important.

Search characteristics: Export

Characteristic	Score	Your home	Sale 1**	Sale 2	Sale 3	Sale 4	Sale 5
Sale Price			$169,000	$165,000	$310,000	$60,000	$68,000
Sale $/sqft			$151	$128	$281	$48	$65
Sale Date			May 2021	Nov 2023	Sep 2022	Feb 2022	Mar 2023
Sale Doc. Num.			2117349145	2333406045	2227625000	2206339368	2310119119
Address		5939 S PEORIA ST	5939 S PEORIA ST	653 W 61ST ST	653 W 60TH ST	619 W 61ST PL	5934 S LOWE AVE HSE
PIN		20-17-406-015	20-17-406-015	20-16-315-085	20-16-309-003	20-16-318-042	20-16-301-039
Neighborhood Code	★★★★	72130	72130	72130	72130	72130	72130
Property Class	★★★	203	203	203	203	203	203
Year Built	★★★★	1888	1888	1878	1898	1888	1898

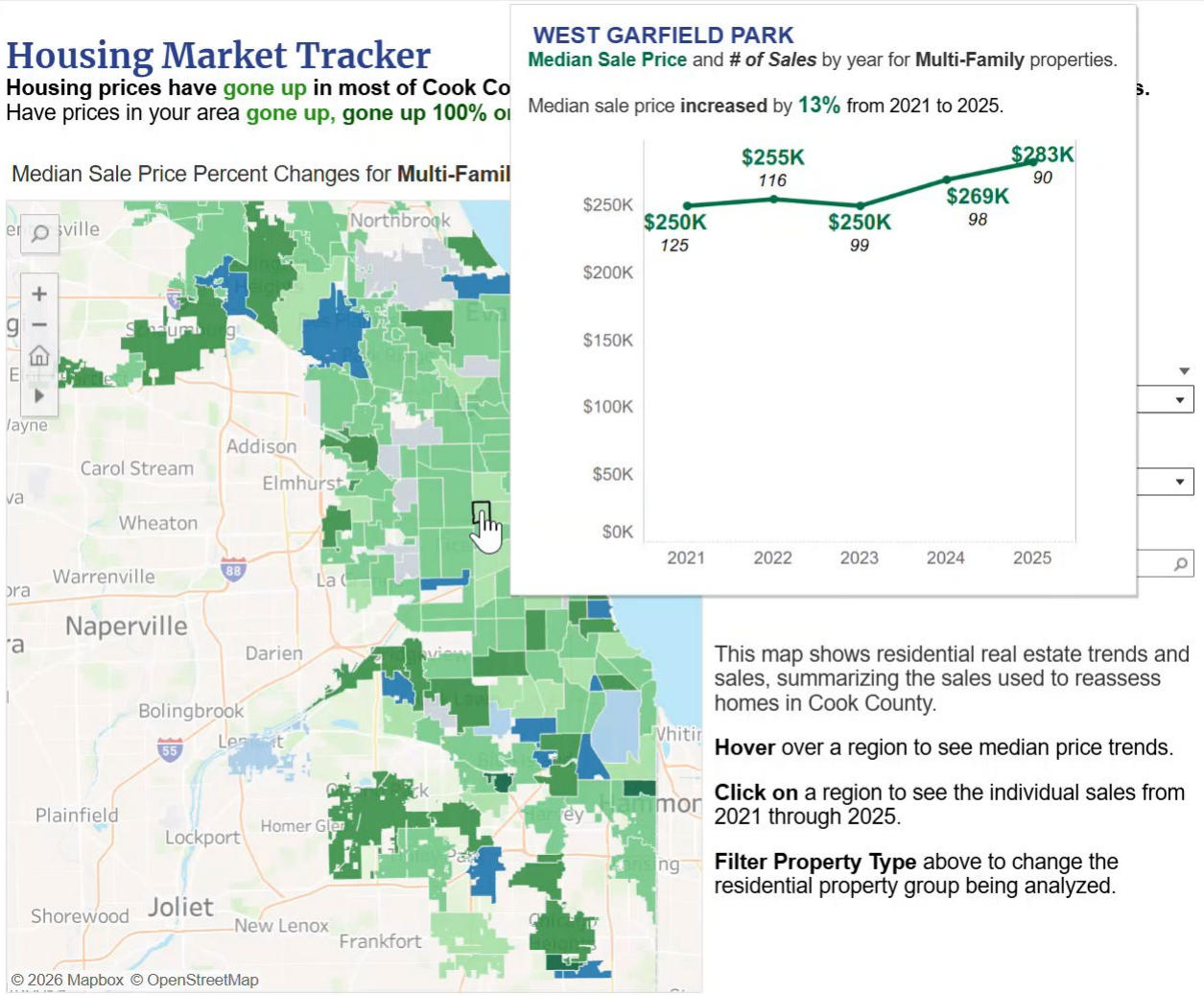
Cook County Housing Market Tracker

- **Uses familiar geographies**
  - Organizes and summarizes sales by Chicago Community Area OR suburban municipality
  - West Garfield Park is a Chicago neighborhood, but it's also a defined geography called a Chicago Community Area



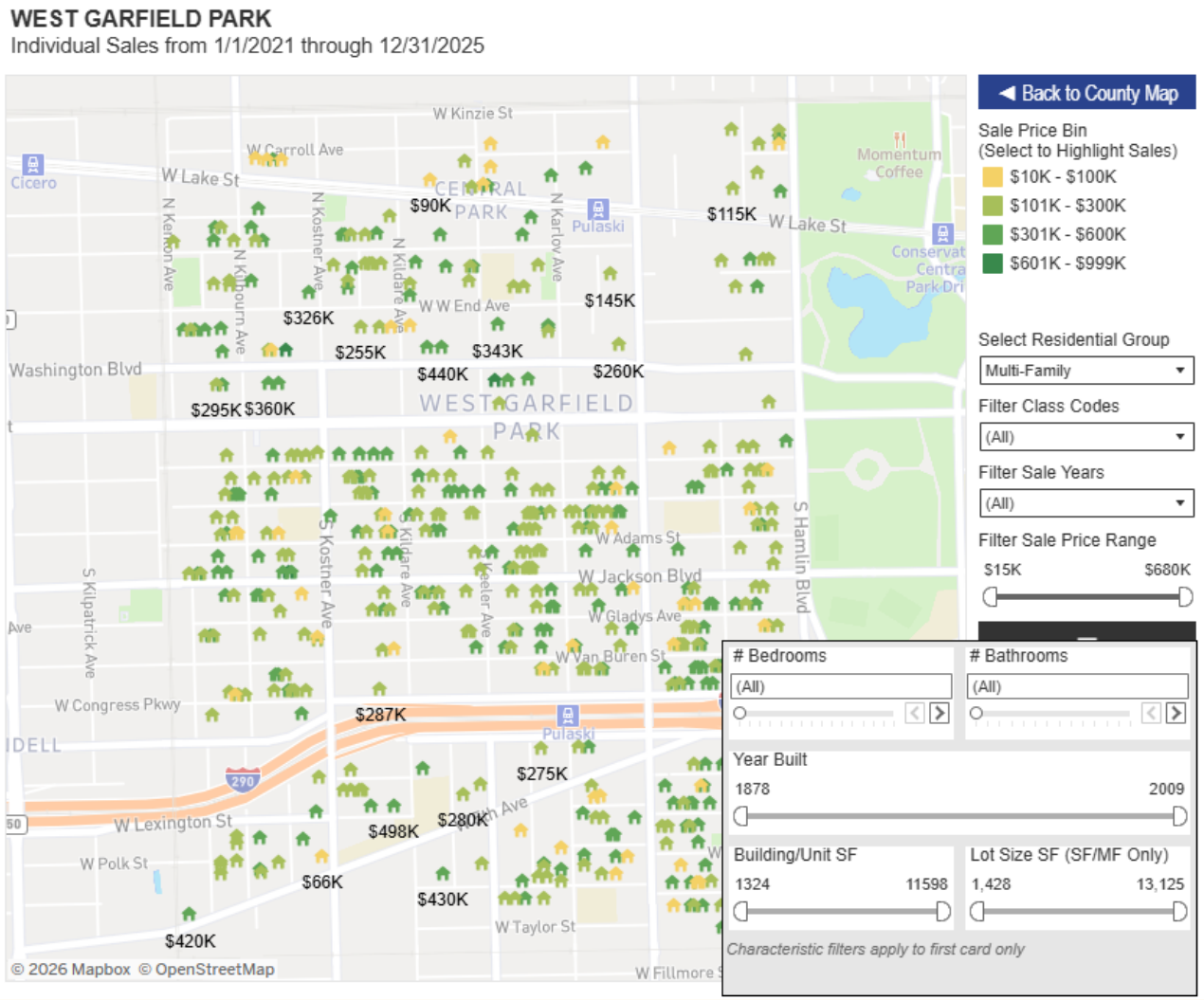
Cook County Housing Market Tracker

- **Allows users to see the sales under the hood**
- **Users can explore the heterogeneity of Cook County**



Cook County Housing Market Tracker

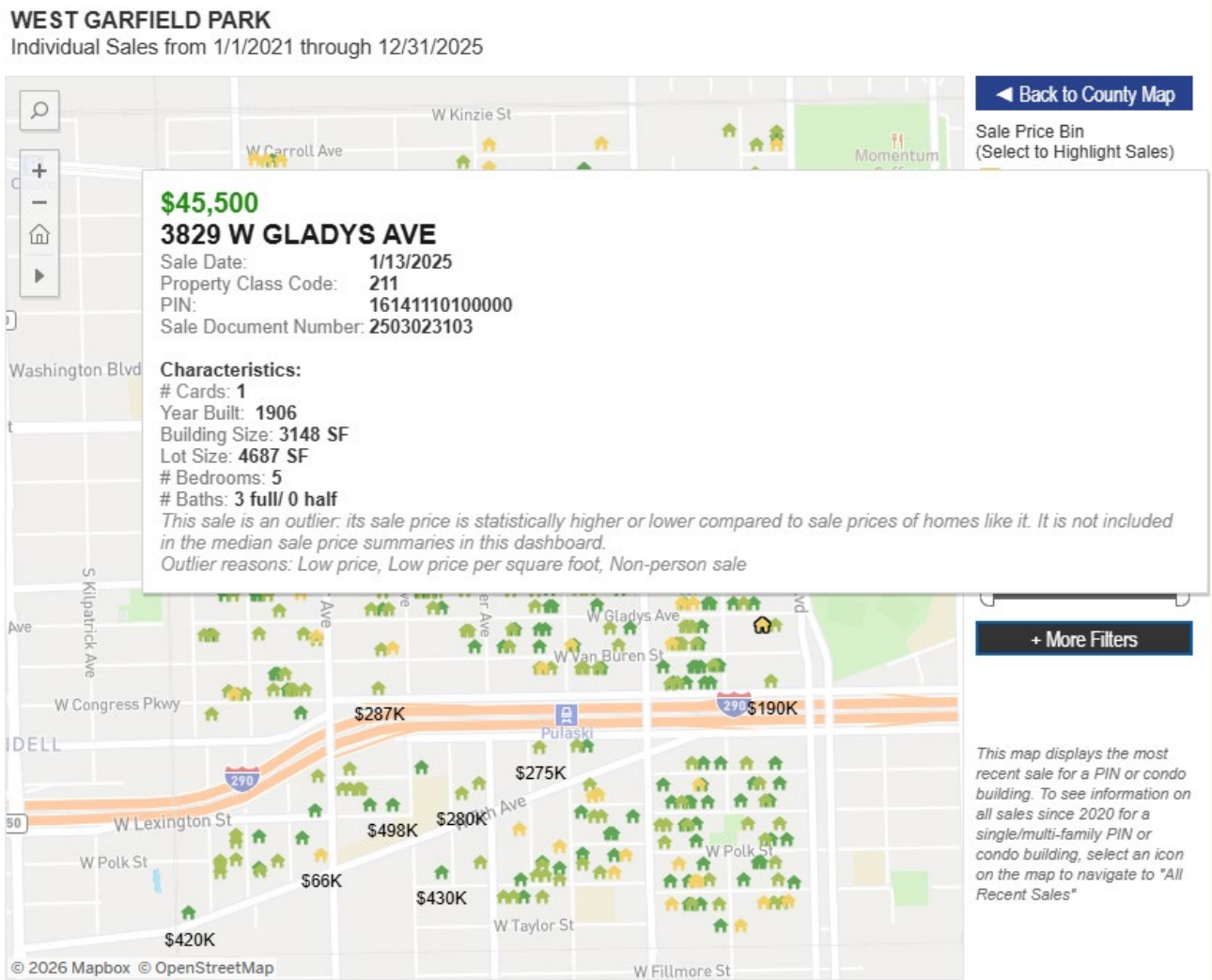
Filter property characteristics, price range, class codes



Cook County Housing Market Tracker

We allow users to see sales excluded from model training and summary data

Notes describe reasons for which the sales validation model flagged it as an outlier



But which sales were used to value *my* home?

To estimate your home's value in a way that fairly and accurately follows the market, the model relies on two kinds of data.

Sales

Statistical models need tens of thousands of sales to calculate price trends.



&

Characteristics

Features of each home and sale *may* be predictive of sale prices.

📐 **Physical**

Building square footage

Year built

🗺️ **Geospatial**

Neighborhood

School district

🕒 **Date of sale**

Sale year, quarter

→

Estimated Value

The model predicts the value of every home being reassessed.



Home Value Report

Characteristics for Top 5 Most Significant Sales

"Characteristics" are data points about a home that help the model compare it to recent sales. The table below shows all of the characteristic that the model used to estimate your home value alongside the characteristics for the top five sales.

Each characteristic has a score indicating how important it was for the model's estimate. One star (★) indicates that a characteristic was not important, while four stars (★★★★) indicates that a characteristic was very important.

Search characteristics:

Export

Characteristic	Score	Your home	Sale 1	Sale 2	Sale 3	Sale 4	Sale 5
Sale Price			$190,000	$152,000	$292,000	$205,000	$280,000
Sale $/sqft			$87	$46	$116	$93	$112
Sale Date			Jan 2022	Aug 2023	Nov 2023	Feb 2022	Oct 2023
Sale Doc. Num.			2201013064	2328410160	2331906033	2206207164	2331113033
Address		4440 W VAN BUREN ST 1FL	4417 W GLADYS AVE 7	4512 W ADAMS ST	4428 W CONGRESS PKY 1	339 S KILBOURN AVE	4446 W ADAMS ST
PIN		16-15-124-026	16-15-124-018	16-15-110-020	16-15-129-030	16-15-124-003	16-15-111-028
Neighborhood Code	★★★★	77102	77102	77102	77102	77102	77102
Property Class	★★★	211	211	211	211	211	211
Year Built	★★★	1908	1888	1893	1903	1908	1898
Building Square Feet	★★★★	3,744	2,196	3,290	2,516	2,209	2,492
Land Square Feet	★★★	3,125	3,125	3,050	3,125	2,725	3,050
Bedrooms	★★	5	4	7	5	6	4
Full Baths	★★★★	3	2	3	2	2	2
Half Baths	★★	0	0	0	0	0	0
Township Code	★★	77	77	77	77	77	77
Number of sales within previous N years of sale/lien date	★★★	0	0	0	0	0	0
Central Air Conditioning	★★★	No Central A/C	No Central A/C	No Central A/C	No Central A/C	No Central A/C	No Central A/C
Apartments	★★★	THREE	TWO	THREE	TWO	TWO	TWO
Attic Finish	★	None	None	None	None	None	None
Attic Type	★	None	None	None	None	None	None
Basement Type	★★	Full	Full	Full	Full	Full	Full
Basement Finish	★★★	Apartment	Unfinished	Apartment	Formal Rec Room	Unfinished	Unfinished
Exterior Wall Material	★★★	Masonry	Masonry	Masonry	Masonry	Masonry	Masonry
Fireplaces	★★	0	0	0	0	0	0
Garage Is Attached	★	No	No	No	No	No	No
Garage Exterior Wall Material	★★	Frame	Frame	Frame	Frame		
Garage Size	★★	2 cars	1 cars	2 cars	1 cars	0 cars	0 cars

+ many more characteristics!

Home Value Report

Summary of the Top 5 Most Significant Sales

The top five most significant sales took place between **2022 and 2023**. These sale prices ranged from **$152,000 to $292,000** and from **$46/sq.ft. to $116/sq.ft.**.

Model Estimate for Your Home

After rounding and other processing, the model's estimate for the value of your home on lien date **January 1st, 2024** was **$270,000** at **$72/sq.ft.**

⚠ **Note**: The estimated value and characteristics in this report may not be identical to what you receive on your Notice of Reassessment. This is because human residential analysts at the Assessor's Office thoroughly review the model's estimates, neighborhood by neighborhood, and may change the property's estimated value and characteristics. To see your home's current assessment and characteristics, visit the **Assessor's website**.



- cookcountyassessoril.gov/home-value-report
- cookcountyassessoril.gov/cook-county-housing-market-tracker

Earned media

CRAIN'S CHICAGO BUSINESS

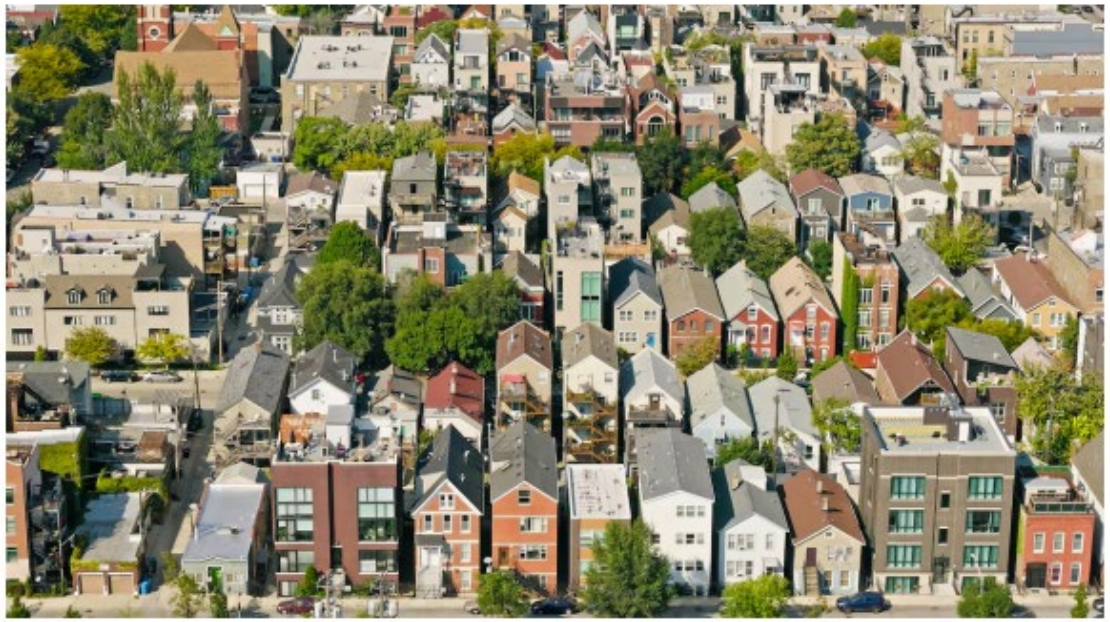
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RESIDENTIAL REAL ESTATE

Cook County has a new home price tracker. Here's how it stacks up.

Gift article | Share | Add as a preferred source on Google



Cook County launches Housing Market Tracker

Daily Herald report **Posted November 14, 2025 7:17 am**

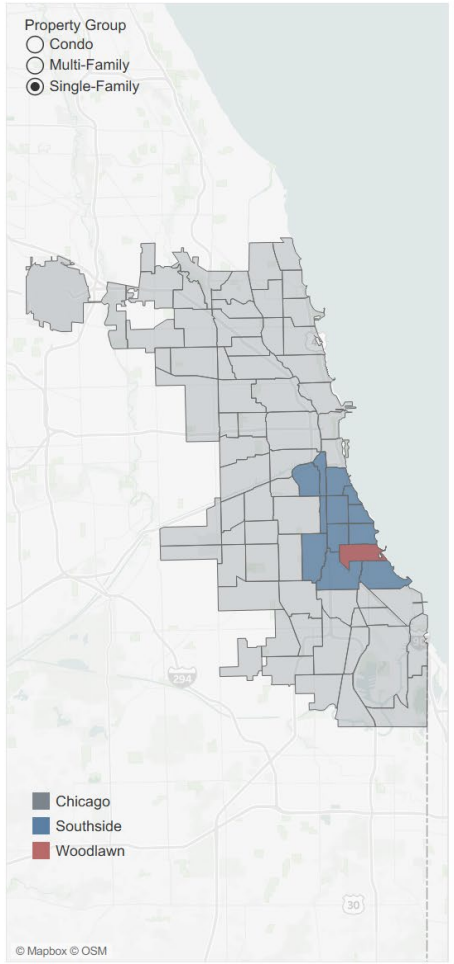
Chicago had some of the worst disparities in Cook County. Businesses' tax bills dropped nearly 20% while residential bills increased by more than 16% during the 2021 to 2023 reassessment cycle. More than 250,000 households saw tax bills jump 25% or more in the same period, according to recently published data from the Cook County assessor's office.

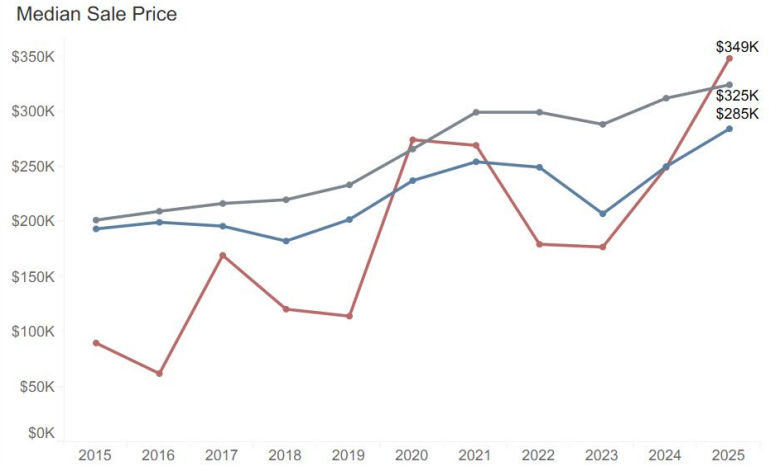
CCAO media channels + outreach

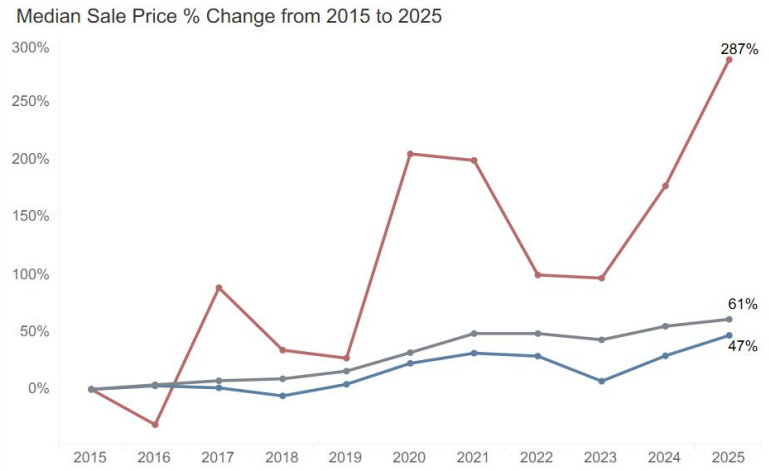
- Community paper press releases and op-eds
- Civic Engagement Department use of Home Value Report
- Data Science Presentation

Annual Reports + Individual Projects









Open Questions

- How do we measure the success of these tools?
- Do people want transparency? If so, what do they use previously inaccessible information for?
- How can this work in other places and times (smaller jurisdictions, periods with a stabler housing market)?